EXPERIMENT : 5

AIM:

Data Collection and Initial Exploration

OBJECTIVE:

To collect, load, and perform initial exploration of the diabetes dataset.

MATERIALS NEEDED:

Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)

Python with libraries: pandas, numpy, matplotlib, seaborn

CODE:

```
import pandas as pd

# Load the dataset
diabetes_df = pd.read_csv('/content/diabets.csv')

# Display the first few rows
print(diabetes_df.head())

# Display information
print(diabetes_df.info())

# Display statistical summary
print(diabetes_df.describe())
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Histograms
diabetes_df.hist(bins=50, figsize=(20, 15))
plt.show()

# Pairplot
sns.pairplot(diabetes_df)
plt.show()
```

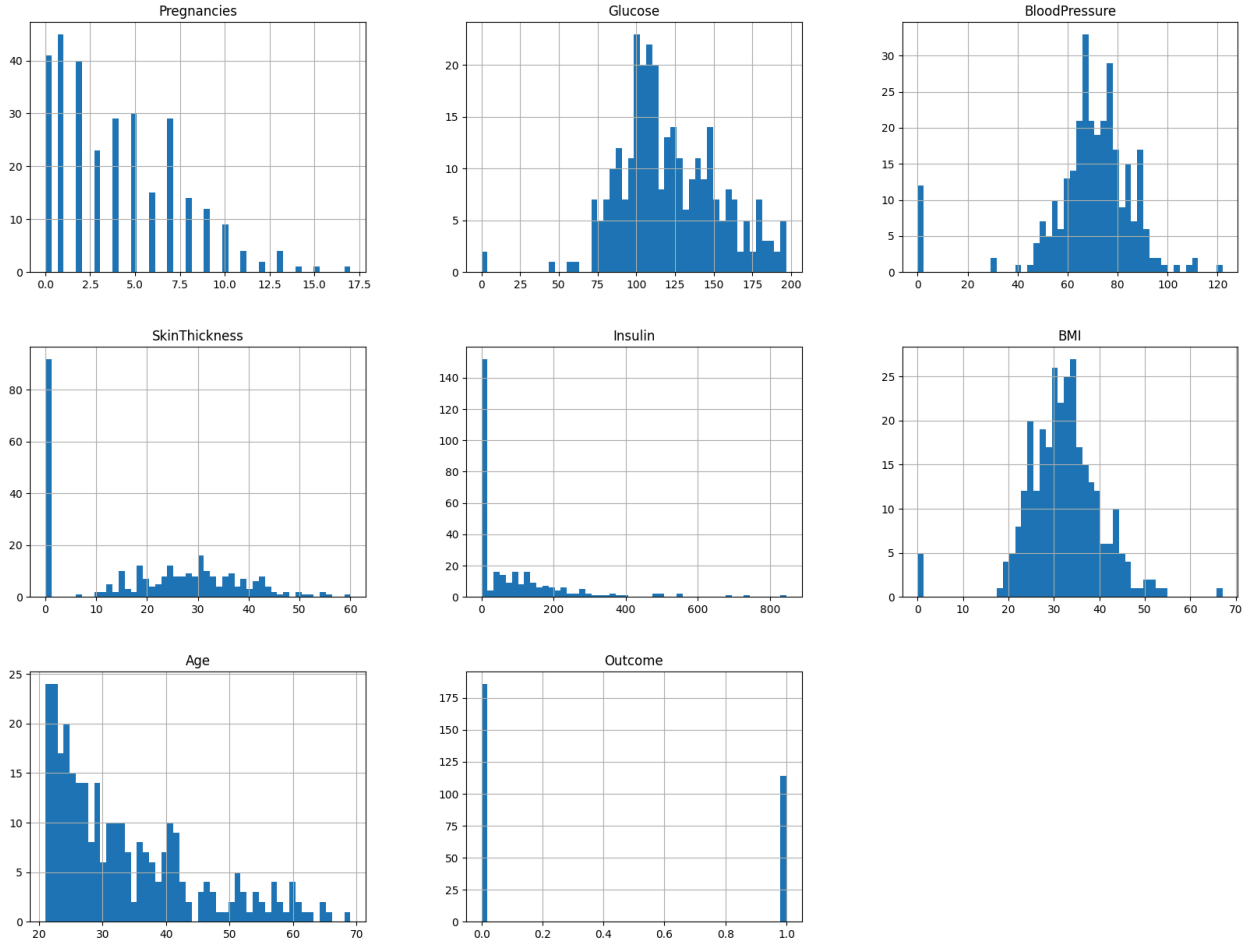
OUTPUT:

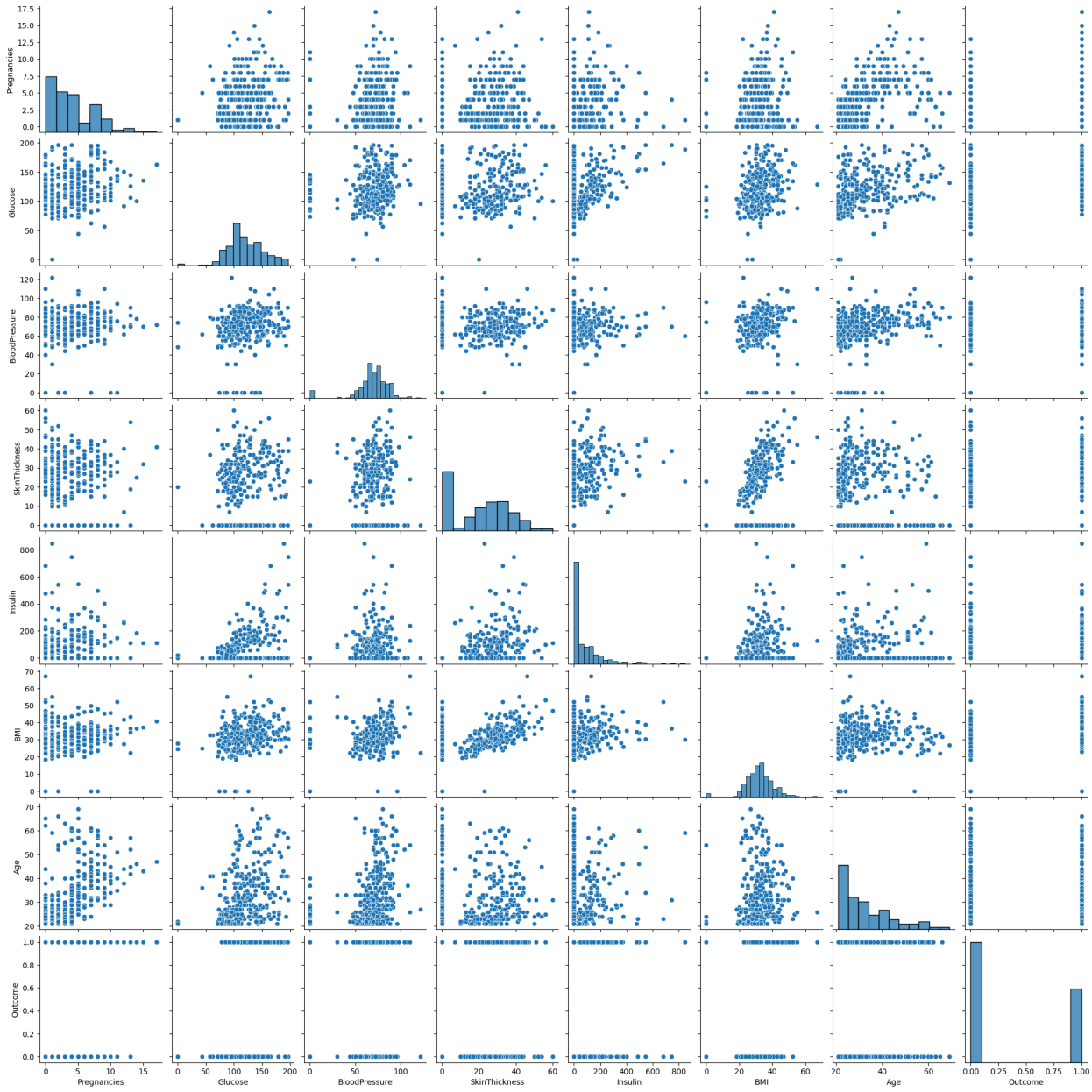
```
...    Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  Age  \
0              6      148             72             35        0  33.6   50
1              1       85             66             29        0  26.6   31
2              8      183             64              0        0  23.3   32
3              1       89             66             23       94  28.1   21
4              0      137             40             35      168  43.1   33

   Outcome
0        1
1        0
2        1
3        0
4        1
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 8 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Pregnancies    300 non-null    int64
 1   Glucose        300 non-null    int64
 2   BloodPressure  300 non-null    int64
 3   SkinThickness  300 non-null    int64
 4   Insulin        300 non-null    int64
 5   BMI            300 non-null    float64
 6   Age            300 non-null    int64
 7   Outcome        300 non-null    int64
dtypes: float64(1), int64(7)
memory usage: 18.9 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   300.000000  300.000000     300.000000      300.00000  300.000000
mean      4.096667  121.096667      69.146667       20.31000   81.300000
std       3.410684   32.091018      18.883160       15.95669  128.271256
min       0.000000    0.000000       0.000000        0.00000    0.000000
25%       1.000000  100.000000      63.500000        0.00000    0.000000
50%       4.000000  115.000000      71.000000       23.00000    0.000000
75%       7.000000  142.250000      80.000000       32.00000  128.250000
max      17.000000  197.000000     122.000000       60.00000  846.000000

              BMI         Age     Outcome
count  300.000000  300.000000  300.000000
mean    32.183667   33.203333    0.380000
std      8.287572   11.385404    0.486197
min      0.000000   21.000000    0.000000
25%     27.600000   24.000000    0.000000
50%     32.050000   29.500000    0.000000
75%     36.625000   40.000000    1.000000
max     67.100000   69.000000    1.000000
```


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
Age
Outcome



RESULT:

The diabetes dataset was successfully collected and loaded using Python. Initial data exploration was performed to understand the dataset's structure, features, data types, and summary statistics.